INCH-POUND
MIL-C-16310D
20 March 1991
SUPERSEDING
MIL-C-16310C
13 December 1967
(See 6.9)

MILITARY SPECIFICATION

CYLINDER COMPRESSED GAS (COMPRESSED AIR AND CARBON DIOXIDE, NONSHATTERABLE AND NONMAGNETIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers special high-pressure cylinders for storage of compressed gases.

1.2 Sizes. The cylinders are of the following sizes, as specified (see 6.2):

Size A - 15-pound capacity CO_2.
Size B - 50-pound capacity CO_2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8120
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SPECIFICATIONS

FEDERAL

A-A-1558 - Paint, Stencil.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-P-15011 - Pallets, Material Handling, Wood Post Construction, 4-Way Entry.
DOD-P-15328 - Primer (Wash), Pretreatment (Formula No. 117 for Metals). (Metric)
MIL-I-17214 - Indicator, Permeability, Low-Mu (Go-No Go).
MIL-V-17360 - Valves, Cylinder, Gas, Carbon Dioxide Fire Extinguisher.
MIL-C-17376 - Caps and Flanges, Compressed-Gas Cylinder, General Specification for.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-101 - Color Code for Pipelines and for Compressed Gas Cylinders.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-278 - Welding and Casting Standard.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

NAVSHIPS 810-1385867 - Decalcomania for Navy Gas Cylinders.

(Application for copies should be addressed to: Commander, Portsmouth Naval Shipyard, Code 202.2, Portsmouth, NH 03801.)

PUBLICATIONS

DEPARTMENT OF TRANSPORTATION (DOT)

49 CFR - Hazardous Material Regulations.

(Application for copies should be addressed to the Superintendent of Documents Government Printing Office, Washington, D.C. 20402.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Cylinders shall be made of corrosion-resistant steel in accordance with any of the analyses shown in table I. Cylinders may be made from pierced billets, cupped plates, or seamless tubing with either welded pressure heads or spun bottoms. When welded cylinders are supplied, the pressure heads used with tubes from a particular heat shall also be manufactured from a single heat in order to simplify gunfire testing.

TABLE I. Chemical composition.

Carbon (max)	Manganese (max)	Phosphorus (max)	Sulfur (max)	Silicon (max)	Molyb (min)	Nickel (min)	Chromium (min)	Titanium
0.12	2.0	0.04	0.04	1.00	----	8.0	17.0	1/
0.08	2.0	0.04	0.04	1.00	----	8.0	17.0	----
0.08	2.0	0.04	0.04	1.00	2.00	10.0	16.0	----

1/ Titanium content shall be not less than 5 times the carbon content.

3.1.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2 Welded cylinders. When welded cylinders are supplied, the joint design, heat treatment, and welding procedure shall be in accordance with MIL-STD-278 (see 6.5). Cylinders shall not be wirebound.

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3.3 General. Cylinders shall be clean and free of scale and weld spatter both inside and out. They shall be of true form and of such shape that they will stand in an upright position. Cylinders shall be nonshatterable and have an average magnetic permeability per unit volume of not more than 2.0, when complete with valve, but without cap. The cylinders shall be equal to or better in strength to corresponding cylinders approved under DOT 49 CFR. The interior and exterior of the cylinders shall be free of defects in accordance with specification 3A of 49 CFR.

3.3.1 Spun bottom cylinders. The sound metal in the bottom of spun cylinders free of nonmetallic inclusions shall be not less than one and one- half times the minimum required cylinder wall thickness. Cylinder bottoms may be drilled at the point of closure and plugged.

3.3.1.1 Leakage. Spun bottom cylinders shall show no leakage when tested with dry air as specified in 4.5.4.

3.4 Pressure rating. The cylinders shall be designed for a working pressure of 2,000 pounds per square inch (lb/in^2) gauge and at test pressure of 3,340 (lb/in^2) gauge. The wall thickness shall keep the maximum wall stress (lb/in^2) at the test pressure, to a value at least 10 percent below the 0.2 percent offset yield strength of the material.

3.5 Valve protection cap and flange. Size B cylinder shall be fitted with a neck ring or cap flange of austenitic corrosion-resisting steel securely held in place on the cylinder, and shall be provided with a valve protection cap in accordance with MIL-C-17376 only when cylinders are fitted with valves.

3.6 Valves. When specified (see 6.2), cylinders shall be fitted with valves in accordance with MIL-V-17360, class b for size A cylinders and class c for size B cylinders.

3.7 Capacity. The minimum capacity of cylinders, at a nominal temperature of 70 degrees Fahrenheit (°F) shall be as specified in table II.

TABLE II. Capacity of cylinders.

Size	Capacity of cylinders (pounds CO_2)	Minimum water volume (cubic inches)
A	15	615
B	50	2300

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3.8 Size and weight. The size and weight shall be as specified in table III.

TABLE III. Size and weight.

Size	Outside diameter	Length 1/	Weight
	(inches)	(inches)	(pounds)
A	6-1/4 ± 1/2	28 ± 1	45
B	8-1/2 ± 1/8	56 ± 1	190

1/ Excluding valve protection cap.
2/ Approximate weight when empty.

3.9 Threading. Cylinders shall be tapped with 1-inch diameter National Gas taper pipe thread, and shall be in accordance with FED-STD-H28.

3.10 Marking.

3.10.1 Cylinders. Each cylinder shall be identified with figures and letters plainly and permanently marked by stamping on the shoulder of each cylinder starting near the neck, as follows (the serial numbers shall be in figures at least 1/4 inch high for size A cylinders and 1/2 inch high for size B cylinders):

One side	Illustration
NONMAGNETIC, the pressure rating and SPUN when spun cylinders are supplied.	NONMAGNETIC 2,000 SPUN
Manufacturer's symbol, manufacturer's serial number, and letters NONSHAT.	MFG D0125 NONSHAT
Government inspector's stamp.	XY
Other side	
Military specification number and date (month and year).	MIL-C-16310 6/67
Original test data (month and year).	6/67

3.10.2 Weight. When cylinders are fitted with valves, the empty and fully charged weights, to the nearest 1/4 pound, shall be indented on the valve body in accordance with MIL-V-17360.

3.11 Painting and identification.

3.11.1 Primer. Cylinders, after being indented with markings specified in 3.10, shall be thoroughly cleaned and then given a coat of primer in accordance with DOD-P-15328.

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3.11.2 Identifying color. Cylinders shall be painted in accordance with MIL-STD-101 except that the letters NON-MAGNETIC shall be stencilled longitudinally above the letters NONSHAT, using the same paint as used for the latter.

3.11.3 Stencilling. In addition to the painting specified in 3.11.1, each cylinder shall be stencilled with the name of the gas contained therein. The stencilling shall be lengthwise on the cylinder on diametrically opposite sides in letters at least 2 inches high. The letters NONSHAT and NONMAGNETIC, at least 1 inch in height, shall be stencilled longitudinally on the cylinder, 90 degrees from the name of the gas. The stencilling paint shall be white in accordance with A-A-1558.

3.11.4 Decalcomanias. Two decalcomanias bearing the name of the gas contained in the cylinder shall be affixed to the shoulder of the cylinder in the same place and between the markings specified in 3.10.1. There shall be no paint over the decalcomanias. Decalcomanias shall be in accordance with Drawing 810-1385867.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Quality conformance inspection. Quality conformance inspection shall consist of the examination and tests of 4.4 and 4.5.

4.3 Sampling for quality conformance.

4.3.1 Inspection lot. All cylinders of the same size presented at one time shall be considered a lot for purposes of quality conformance inspection.

4.3.2 Sampling for examination. A random sample of cylinders shall be selected in accordance with table IV from each inspection lot of material for examination of visual, dimensional, and weight characteristics (see 6.6).

TABLE IV. Sampling for visual, dimensional and weight examination.

Number of cylinders in an inspection lot	Number of cylinders in sample
15 and under	7
16 to 40	15
41 to 110	25
111 to 300	35
301 to 500	50
501 to 800	75
801 to 1300	110
1301 to 3200	150
3201 and over	225

4.3.3 Sampling for gunfire tests. Three cylinders representing each lot of steel entering into the manufacture of the cylinders on a contract or order shall be selected for the gunfire tests specified in 4.5.1. If the contract or order calls for less than 30 cylinders, only one sample cylinder is required.

4.4 Visual, dimensional, and weight examination. Each of the sample cylinders selected in accordance with table IV shall be visually and dimensionally examined and weighted to verify compliance with this specification. Any cylinder in the sample containing one or more visual or dimensional defects or over required weight shall be rejected (see 6.6).

4.5 Tests.

4.5.1 Gunfire test. Cylinders for gunfire test shall be selected in accordance with 4.2.3 and marked "For test" (see 6.2). The cylinder shall be fitted by the contractor with a satisfactory valve which can be opened and

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closed to check the pressure. The cylinders will be charged with air at the Naval Weapons Laboratory to a pressure of 2,000 lb/in^2 gauge. While under this pressure, the cylinder will be hit perpen-dicular to the longitudinal axis with a 40-mm armor piercing projectile at a velocity of 1,400 feet per second. Under this test, the cylinder wall shall remain in one piece, showing no fragmentation at any point of tear other than that generally encountered at the projectile entrance and exit of the cylinder (see 6.7).

4.5.2 Leakage test. If furnished with valve and charge, each cylinder selected in accordance with 4.3.2 shall be tested for leakage by submerging it in water at a temperature of not less than 65°F for a period of not less than 2 minutes. Any evidence of leakage shall be cause for rejection. When valves are not purchased with the cylinders, then this test shall be cause for rejection. When valves are not purchased with the cylinders, then this test shall be performed on all cylinders using a minimum number of suitable valves loaned by the manufacturer. If so desired, a test method utilizing a halide gas and halide leak detector, and approved by the contracting activity concerned, may be substituted for the above water submergence test.

4.5.3 Uniformity. Where spun bottom cylinders are supplied, careful and frequent observation shall be made to assure that uniformity of the spinning and welding processes is maintained. From each sample group of cylinders selected in accordance with table IV, another selection of 1 percent of the sample shall be made. The cylinders selected shall be sectioned across the spun bottom and examined to determine conformance with 3.3.1.

4.5.4 Dry air tests. Prior to heat treatment, the bottoms of spun cylinders shall be subjected to a dry air test pressure of not less than 2,000 lb/in^2 gauge for 1 minute with the inverted bottom filled with water.

4.5.5 Permeability test. Each cylinder selected in accordance with 4.3.2 shall be tested for magnetic properties. The magnetic permeability shall be determined by means of a permeability indicator of the go-no go type conforming to or equal to that specified in MIL-I-17214. The magnetic permeability shall not exceed 2.0 after completion of fabrication.

4.5.6 Hydrostatic test. Each cylinder shall be hydrostatically tested as required by specification 3A of 49 CFR.

4.6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Packing. Cylinders shall be packed for shipment in accordance with level A, B, or C as specified (see 6.2).

5.1.1 Size A cylinders. Size A cylinders shall be packed as specified in 5.1.1.1 through 5.1.1.3 (see 6.2).

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5.1.1.1 Level A. Size A cylinders shall be packed in boxes in accordance with PPP-B-585, PPP-B-601 or PPP-B-621 for overseas shipment. The gross weight shall not exceed 200 pounds.

5.1.1.2 Level B. Size A cylinders shall be packed in boxes in accordance with PPP-B-585, PPP-B-601 or PPP-B-621 for overseas shipment. The gross weight shall not exceed 200 pounds.

5.1.1.2.1 Level B alternative. As an alternate method, size A cylinders may be packed on pallets in accordance with MIL-P-15011. The cylinders shall be secured to the pallet with coated or galvanized 1-1/4- by 0.035-inch flat steel strapping, or equivalent round steel wire. Additional wood bracing shall be provided as necessary to adequately secure the load. The gross weight shall not exceed 2100 pounds.

5.1.1.3 Level C. The cylinders shall be prepared for shipment in a manner which will insure acceptance by common carrier and safe arrival at destination.

5.1.2 Size B cylinders. Unless otherwise specified (see 6.2), size B cylinders shall be packed in pallets in accordance with MIL-P-15011. Loading, strapping, and wood collar details shall be as shown on figure 1. The strapping shall be coated or galvanized. The gross weight shall not exceed 2100 pounds.

5.2 Marking. In addition to any special marking required in the contract or order, marking for shipment shall be in accordance with MIL-STD-129. Nomenclature shall be the exact nomenclature for the part of item that has been approved under the contract or order.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The intended use for cylinders covered by this specification is in applications requiring the nonmagnetic characteristic. For such applications the DOT Specification 3AA, steel cylinders in accordance with RR-C-901 would not be suitable. Each application of cylinders as specified herein should be evaluated because DOT Specification 3AL, aluminum cylinders covered by RR-C-901 might be suitable.

6.1.1 Cylinder sizes. The cylinder sizes covered by this specification are for use as follows:

(a) Size A, 15-pound capacity cylinders - For use as portable fire extinguishers (see MIL-E-24269).
(b) Size B, 50-pound capacity cylinders - For use in hose reel or fixed, built-in fire extinguisher systems (see MIL-E-2185 and MIL-E-2186).

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6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Size of cylinder required (see 1.2 and 6.1.1).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).
(d) When valves are to be shipped with cylinders, and type of valve (see 3.6).
(e) That for Naval purchases the cylinders for gunfire tests will be shipped to the Naval Weapons Laboratory, Dahlgren, VA 22448, at the contractor's expense; that the gunfire test will be made at the contractor's expense; and that if the contractor desires return of the cylinders, they will be returned at his expense (see 4.5.1).
(f) For size A cylinders, level of packing (see 5.1 and 5.1.1).
(g) For size B cylinders, level of packing if other than as specified (see 5.1 and 5.1.2).
(h) When condition of cylinders is other than as specified (see 6.3).

6.3 Condition of cylinders. Unless otherwise specified (see 6.2), cylinders ordered in accordance with this specification are empty and contracting activity without valves.

6.4 For Naval purchases. The Department of the Navy will arrange to conduct a reasonable number of preliminary gunfire tests, entirely at the expense of the exhibitors, so that prospective bidders may obtain data about the necessary characteristics of the material of cylinders which will enable the cylinders to pass the gunfire test. Such tests will be considered to be information for the bidders only. Regardless of the outcome it will be necessary for the requirements specified in 4.3.3 and 4.5.1 to be fulfilled upon the award of any contract.

6.5 Welded cylinders. When welded cylinders are acquired by a Naval agency or for ultimate use by the Navy, approval of design and construction details should be obtained from the Naval Sea Systems Command, Department of the Navy, Washington, D.C. 20362-5101.

6.6 Lot acceptance and rejection criteria. Lot acceptance and rejection for the visual, dimensional, and weight examination should be in accordance with table V. Acceptable quality level (AQL) is approximately 1.5 percent defective.

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TABLE V. Examination acceptance and rejection criteria.

Number of cylinders in an inspection lot	Number of cylinders in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	7	0	1
16 to 40	15	0	1
41 to 110	25	1	2
111 to 300	35	1	2
301 to 500	50	2	3
501 to 800	75	3	4
801 to 1300	110	4	5
1301 to 3200	150	5	6
3201 and over	225	8	9

6.7 Gunfire test. If any cylinder fails to pass this test (see 4.5.1), the contractor will be permitted to reheat-treat all the cylinders manufactured from the same heat of steel, and additional cylinders should be selected as specified in 4.3.3. If the cylinders pass this retest, all cylinders made from the same heat, provided they receive the same heat treatment, will be accepted. Failure of any cylinder in the retest will cause rejection of all cylinders made from that heat.

6.8 Subject term (key word) listing.

Cupped plates
Fire extinguishers, portable
Fire extinguishers, hose reel or fixed built-in
Pierced billets
Spun bottom

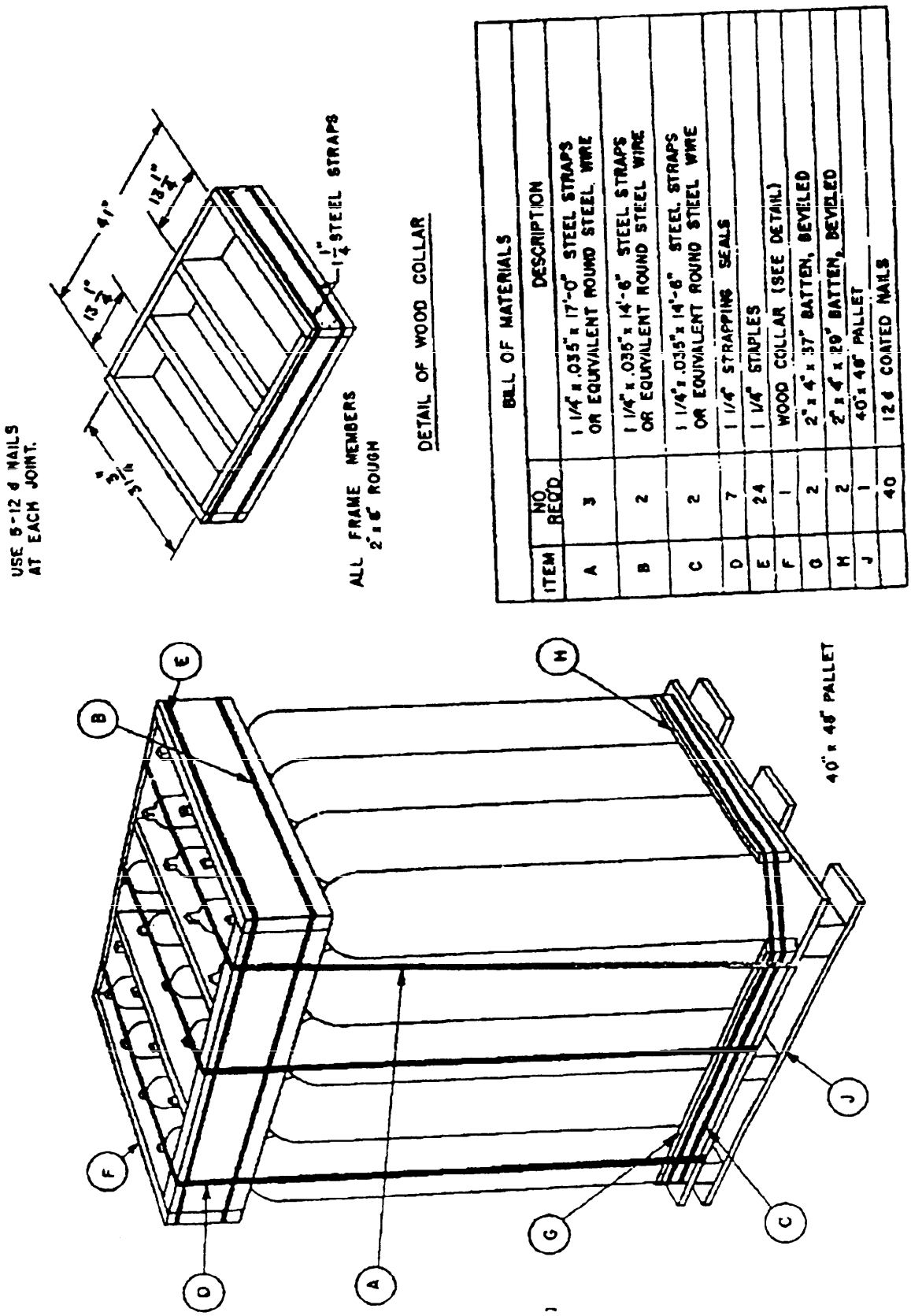
6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - SH
Air Force - 68

Review activities:
Army - ME
Navy - SH, YD
Air Force - 68

Preparing activity:
Navy - SH
(Project 8120-1002)

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BILL OF MATERIALS

ITEM	NO REQD	DESCRIPTION
A	3	1 1/4" x .035" x 17'-0" STEEL STRAPS OR EQUIVALENT ROUND STEEL WIRE
B	2	1 1/4" x .035" x 14'-6" STEEL STRAPS OR EQUIVALENT ROUND STEEL WIRE
C	2	1 1/4" x .035" x 14'-6" STEEL STRAPS OR EQUIVALENT ROUND STEEL WIRE
D	7	1 1/4" STRAPPING SEALS
E	24	1 1/4" STAPLES
F	1	WOOD COLLAR (SEE DETAIL)
G	2	2" x 4" x 37" BATTEN, BEVELED
H	2	2" x 4" x 29" BATTEN, BEVELED
J	1	40" x 48" PALLET
	40	12 d COATED NAILS

FIGURE 1. *Unit load for cylinders.*

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-16310D	2. DOCUMENT DATE (YYMMDD) 20 March 1991

3. DOCUMENT TITLE
Cylinder Compressed Gas (Compressed Air & Carbon Dioxide, Nonshatterable)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Commander Naval Sea Systems Command (5523)	b. TELEPHONE (Include Area Code) (1) Commercial: (703) 602-5585 (2) AUTOVON: 332-5585
c. ADDRESS (Include Zip Code) Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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